

CS 112

Introduction to Data Structures

WEEK 04

Vectors — Dynamic Arrays

Eric Araújo

Calvin University · Fall 2026

This Week

- 1 Why a plain array isn't enough
- 2 `new` and `delete`
- 3 The stack and the heap
- 4 Building a dynamic array
- 5 Growing when full
- 6 Destructors — now you need one
- 7 Copy constructors: shallow vs deep **KEY**

Arrays: The Good and the Bad

Good

- Simple to declare
- Indexing is fast — one multiply and add
- Contiguous, so the cache loves them

Bad

- Size fixed at **compile time**
- Can't grow when you need one more
- No idea how many slots are actually used

Python's `list` grows on demand. This week we build that — and find out what it costs.

Memory You Ask For

```
int *arr = new int[capacity]; // ask the OS for space

arr[0] = 42;                  // use it like an array

delete [] arr;                // give it back
arr = nullptr;                // don't keep the address
```

- `new` returns the **address** of the block
- The size can be a variable — decided while the program runs
- `delete []` for arrays, plain `delete` for single objects

Every `new` needs exactly one matching `delete`. Forget it and you leak; do it twice and you corrupt the heap.

Stack and Heap

Stack

- Locals and parameters
- Freed automatically at end of scope
- Size known at compile time
- Small and fast

Heap

- Whatever `new` hands you
- Freed only when *you* say so
- Size decided at runtime
- Large, slightly slower

stack	Vec v	myArray = 0x5f2a10
heap	0x5f2a10	[3 7 11 _ _]

The object lives on the stack; the elements live on the heap. That split is the whole design.

A Dynamic Array

```
class Vec {
public:
    Vec();
    Vec(const Vec &original);           // copy constructor
    ~Vec();                             // destructor
    unsigned getSize() const;
    void append(const Item &it);
    Item& operator[](unsigned i);
private:
    Item *myArray;                      // heap block
    unsigned mySize;                    // slots in use
    unsigned myCapacity;                // slots available
};
```

size is how many items you have. **capacity** is how many fit before it must grow. Keeping them apart is what makes appending cheap most of the time.

Appending When There's Room

```
void Vec::append(const Item &it) {  
    myArray[mySize] = it;  
    ++mySize;  
}
```

before	[3 7 _ _]	size 2, cap 4
after	[3 7 11 _]	size 3, cap 4

Two operations, no matter how big the array is. **Constant time.**

Appending When It's Full

```
void Vec::append(const Item &it) {
    if (mySize == myCapacity) {
        unsigned cap = (myCapacity == 0) ? 1 : myCapacity*2;
        Item *bigger = new Item[cap];
        for (unsigned i = 0; i < mySize; ++i) {
            bigger[i] = myArray[i];           // copy everything
        }
        delete [] myArray;                   // release the old
        myArray = bigger;
        myCapacity = cap;
    }
    myArray[mySize] = it;
    ++mySize;
}
```

Now it copies all n items: **linear time**. Doubling means this happens rarely — more on that next week.

Why double the capacity instead of adding one slot?

- A.** Doubling uses less memory
- B.** Adding one would make every append copy the whole array
- C.** The heap only hands out sizes that are powers of two
- D.** It doesn't matter — both are equally fast

Now You Need a Destructor

```
Vec::~Vec() {  
    delete [] myArray;  
    myArray = nullptr;  
}
```

C++ writes a destructor for you — but it only destroys the *members*. It destroys the pointer, not what the pointer points at.

The rule: if your class calls `new`, your class must define a destructor. Otherwise every `Vec` that dies leaves its heap block behind.

When does a destructor run?

- A.** Only when you write `delete`
- B.** When a local object reaches the end of its scope, or is deleted
- C.** At the end of `main()`, for every object
- D.** When the garbage collector decides

The Copy That Bites

```
Vec a;  
a.append(3);  
a.append(7);  
  
Vec b = a;           // what exactly got copied?
```

a.myArray	0x5f2a10	→ [3 7]
b.myArray	0x5f2a10	→ the same block!

The default copy duplicates the **pointer**, not the data. Two objects now own one block — a **shallow copy**.

What goes wrong with that shallow copy?

- A.** Nothing — sharing the block saves memory
- B.** Changing `b[0]` also changes `a[0]`
- C.** Both destructors delete the same block — undefined behaviour
- D.** Both B and C

Deep Copy: the Copy Constructor

```
Vec::Vec(const Vec &orig) {  
    mySize      = orig.mySize;  
    myCapacity  = orig.myCapacity;  
    myArray     = new Item[myCapacity];    // our OWN block  
    for (unsigned i = 0; i < mySize; ++i) {  
        myArray[i] = orig.myArray[i];    // copy the values  
    }  
}
```

Same shape as the growth code: allocate, copy, own. Now a and b are independent, and each destructor frees its own block.

Three Times It Copies

```
Vec b = a;                // 1. explicit copy

void show(Vec v);         // 2. pass-by-value
show(a);

Vec makeVec() {           // 3. return by value
    Vec local;
    return local;
}
```

Case 2 is why week 1 said *pass big objects by const reference*. Every by-value call to `show()` copies the whole heap block.

Demo



Watching a shallow copy destroy itself.

(double free detected in tcache 2)

Returning a Reference: operator[]

```
Item& Vec::operator[](unsigned i) {  
    return myArray[i];           // a reference, not a copy  
}  
  
v[2] = 99;           // works because v[2] IS the slot
```

Return Item and you get a copy — assigning to it would change a temporary and vanish. Returning Item& makes the call an *alias* for the element.

The this Pointer

Python

```
def set_first(self, v):  
    self.my_first = v
```

C++

```
void setFirst(Item v) {  
    this->myFirst = v;  
}
```

C++ passes the object silently as `this` — a pointer to the object the method was called on. You rarely write it, but it's there, and `this->x` is just `(*this).x`.

Week 04 Recap

- `new` gets heap memory at runtime; every `new` needs one `delete`
- **size** vs **capacity** is what makes append cheap most of the time
- Growing means allocate, copy, delete, repoint — linear, so we double
- If your class calls `new`, it needs a **destructor**
- The default copy is **shallow** — aliasing plus a double delete
- Write a copy constructor that allocates its own block

Next: a08 · a09 →

This deck stands on earlier CS112 materials by
Joel Adams and **Victor Norman** · adapted and extended by **Eric Araújo**

